# ActInf Textbook Group ~ Cohort 7 ~ Session 5 (Chapter 2 +++ Open Discus

*TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 5 (Chapter 2 +++ Open Discussion) ~ 8/12/2024 | 47:02*

all right we're in an open slen session
so Adam perhaps begin with the question
and then Andrew go for it okay thank you
so in chapter
seven in the summary it says that
um inference using both simple and more
complex generative models can always
proceed through free energy
minimization which illustrates the
generality of the approach
and my question is about uh whether or
not there is a partition function for
the model of the minimum free energy
State
uh and I wanted to note how
about in section 7.3 they talk about
sorting the states of the world into
categories of whether the could treater
has control or whether it does not and
then it says that the the transition
prob ilities for the ones that it
doesn't are invariant and then each
state that the creature has a control
over the world has like a unique
transition probability so then it's
always going to try to select the
transition
probability which is most effective at
bringing it towards a free energy
minimum state
right so
um that would be the
equilibrium I suppose which was making
me think about um whether or not there's
a partition

function yeah so uh it's kind of like to answer that there are a few different pieces going on so yeah with like you can see in that chapter where um let's see
page 126 the the just the second page of the chapter you can see like like the breakdown of the the B Matrix
as they call it at the bottom of the page and so like yeah you're going to have a unique B Matrix for uh every hidden State Factor um and it's the dimensionality is kind of like um the we're gonna what is the state going to be at the next time step conditioned on uh what you infer the state to be in the current time step as well as um a a policy so it's kind of like the agent will uh you know they'll try to think what can I do now to change what I think is going on now in terms of the Hidden state to um you know a particular hidden state that I I kind of want to see in the next step and I I would say in a kind of all things controlled for kind of way what you said is correct it it will you know go with what the action that is you know most
probable uh to get it to a state it wants to be in and that said because free energy is being run across all of these different you know components of the model um it's not necessarily the case that you can just look at the B Matrix and immediately know what the agent is going to do just because one state trans position is more probable than another um when it's Computing

expected free energy for example that
you know gets broken down into epistemic
and pragmatic value um an agent might
choose to transition it might choose to
do something to make a state transition
that it isn't really clear on for
example it might compute like oh there's
epistemic value there um you know and
then if you have your agent like
learning and B Matrix those values will
get updated over time um so I I hope
that that kind of framing is is useful
in some way as far as a partition
function um so I just want to I want to
better understand and maybe it'll be
helpful for others too how how would you
define a partition function in in the
way that you mean
it yeah you know what um I don't even
really know like honestly I've never
actually formally studied um statistical
physics
so when I like I've asked there's also
like a partition function
in is it number Theory and I was I asked
all over nil about it on on YouTube and
he gave me this I'm gonna be rambling on
he he sent me a paper that was actually
discussing like the relationship between
the partition function in number Theory
with the partition function in
statistical physics in my question to
him was was it the same one and he said
he didn't really know and I had the
paper I was reading it was interesting I
forget I don't have it off hand and um
in this case I'm kind of wondering like
the word partitioning in terms of

sorting the two states of the world from
the ones we can change the ones that we
cannot and then at equilibrium like the
um those States become stationary I
suppose in some sense and uh so but I'm
also wondering
I'm also wondering is that would the
partition function manifest as the
different aspects of active inference
that are supposedly emerging
like let me give a few few thoughts on
this just really briefly because you're
you're in the right track partitions
being used in a um hate to reuse the pun
but it's a particular way and so a lot
of the momentum in the current kind of
wave of active inference is from a 2019
paper from Carl friston so a free energy
principle for a particular physics
particular here's the joke it Cleaves
off particles we call the entity a
cognitive particle whether it's rock
metronome or like more sophisticated Etc
and then it's a physics for those kinds
of partitioned systems so that is taking
a basian graph and then doing certain
partitions on the graph there's other
uses for the term partition like you
brought up the term from stat Mech so
here partition function describes
statistical properties of a system in
thermodynamic equilibrium so there the
free energy is like Gibbs free energy
like why do candles burn once they're
activated forward like why is time going
that way why don't candles like unburn
so that's on the
thermodynamics then over the last like

00 years all of these Mega developments
with information and thermog Dynamics
like analogies between them or
isomorphisms but also Unity between them
like with the land hour limit and how
much energy does it take to read or
write a bit so it's a kind of info
thermodynamic bace that the free energy
principle is related to um so are there
well-described summary variables that
describe systems that are in different
kinds of info dnamic
equilibrium there are so the simpler
case is like a fixed points so that be
like the rock is just resting on the
ground it's like the belief is not
updating then the next step would be
like the simple update like this bilard
ball hit the rock it moved this much
that's like this data point hit the
prior it moved that much and then
there's all kinds of like steady states
that could be like an
oscillator so you're going the right
area to see how those things are
formalized
yeah that um so I guess it
isn't like it wouldn't be describing
because I did read on on YouTube there
it was saying the thermodynamic
properties of a system like the
temperature and
volume and
uh so if in an in what would those be um
parallel to like in an active inference
system would it be the aspects like
decision making and balancing
exploration and

learning great question Andrew what
would you say to
that
um I mean yeah
it's I'm trying to figure out how how to
phrase that well I mean it's you'd want
to look at the I would just refer to
like the the breakdowns of variational
and expected free energy I suppose it's
it's it's expected free energy where
you're going to see that as I mentioned
earlier that epistemic and pragmatic
value um
differentiation and that that plays out
in learning as well like expect like
free energy is being minimized during
the process of of of learning where
rather than you know
inference in the context of of active
inference like inference is inferring
states of the world uh learning is
inferring uh sort of model like
parameters values right so you're you're
going to be like that the agent learns
to adjust its own parameter values it's
also going to infer Moment by moment
States um ex uh the pragmatic and osmic
value play
into that overall process and and I mean
all these things are kind of happening
um s mously so so so for me it's it's a
little tough for me to like kind
of rip out one part of that process and
say that one thing is like what's
happening um but I'm not sure if I'm
answering the question super well or not
that's that's I think that was a key
piece which is the epistemic and

pragmatic value are a way to like
reweight the policy prior into the
policy post interior so for a kind of
thing that's not doing the cognitive
task of something like policy updating
so if something's drawing from habit
that could be like a go to the simplest
and fixed habit it's like the rock is
always in location one so it's like it's
being sampled from location one with
100% again and again it's kind of like a
redundant or a tot logical statistical
distribution and that is also what comes
up is like def defining things as
they're measured to be so then rather
than the the space of possibly
dissipative things you say like well the
measurements over this time scale are
such that it is returning to an
attractor and then what are the Dynamics
of that attractor is it like just a
marble going to the bottom of a bowl and
perturbation brings it back or does the
bottom of the bowl adjust that's like
learning is there another landscape then
all all those kinds of questions
you think it'd be helpful to thank you I
don't know
there's I like the
um one of one of the simpler equations
from active inference for just like
Computing like the final posterior over
policies I I just I've always found that
interesting and like helpful for like
describing to folks lesser familiar with
active inference like like your your
posterior over policies typically is
like um a softmax fun softmax function

over negative expected free energy times
gamma which is referred to it's like a
um Precision over actions and then
um minus
um I'd rather pull it up actually just
so I don't miss speak here the the main
point is that it includes expected free
energy and variational free energy and
habits all of which you know Daniel just
mentioned a moment ago and it helps to
give a an intuitive sense of like all
three of these components are are are
going into the agent's Behavior like
what action they end up selecting right
it's like what expect expected free
energy as what it expects to happen and
that's where it's uh prior over
observations come in so it's that's
where its preferences come in and then
um variational free energy is just
reducing kind of uncertainty in the
moment and then habits are like these
empirical priers that have been built up
over
time
yeah another kind of
big moment in helping think about this
was this particular
kinds showing continua from like in
continua of modeling continuity not as
much as like an ontological Claim about
any actual system but in terms of a
modeling continuity with systems that
are being interfaced with differently
ranging from kind of like inert
unidirectional cause to kind of simpler
inert to simpler nonplanning like on
through like planning like with other

cognitive phenomena like
reflexivity meaning creation things like
that so the simple ones really do look
like this the graphs but this graph on
the bottom is way more of like a
schematic and in play those graphs might
be very large
um sh Harsha or Courtney have any
question or like different direction to
explore I don't at the moment I was just
kind
of listening to your guys conversation
and seeing thank you C from that since
I'm not that far along yet cool thank
you yeah it's like always like skipping
around on on a record here's one this is
the basian
mechanics
paper this is kind of a summary slide
thank you
shha so this paper this is like docon
stiva devel and Maxwell ramstead and
others working on the basian mechanics
so here's classical physics classical
mechanics stationary action that's like
why the baseball goes in the parabola
then statistical physics says well it's
not just like planets and baseballs and
gravity it's other certain behaved
summary statistics like temperature and
pressure that was what we were just
looking at with the partition
functions um
Quantum moves that into the not just the
scale that was being explored but also
like the imaginary or like axis that's
orthagonal so that's why there's the
imaginary numbers and the rotation and

it's an equivalent physics of Uncertain
systems especially interpreted in this
kind of scale-free way so it's not just
like unique mechanisms related to
electrons but those contribute to it and
and uh constitute why the variability
patterns are the way they are but people
have applied Quantum formalisms across
different systems even without relying
on like some sort of um Reliance on
electrons or protons or anything and
then kind of in that same frame and
generalization and you can generalize
Quantum to be statistical physics with
no um wave distribution and you could
summarize statistical physics to be like
that's why the baseballs um moves like a
parabola instead of like
diffusing basian physics extends that to
arbitrary State space so quantum
mechanical State spaces maps of quantum
mechanical or Stat or classical are
certain within certain categories of
maps and
models so this basing mechanics thread
especially a lot of the rigor that
Dalton and Chris fields and some others
brought in is only in the last like two
post again post 2019 prison kind of
entry into that fundamental uh level of
the interface but it's kind of also
threaded going prior but took a lot of a
a consolidation and development turn and
this is not brought up in the 2022
textbook so it's kind of like the
textbook is at this
checkpoint where the like the Baseline
the so in the textbook it's like what

are the math
areas and it says linear algebra so
matrix
multiplication and then dynamical and
stochastic systems like probability on
dynamical
systems that is like a checkpoint and
then that's been some of the most
interest over the last years was what
what is the basis of basy
mechanics how does it it matter or
differ from the um 2022 textbook Style
Andrew anyone can give a thought or a
different question
yeah sorry Dana what what were you
refering to in terms of the linear
algebra and the stochastic
probability
um but in in the textbook so one part is
where they sort of uh mention the kinds
of maths that are needed like those are
good keywords to
see like here linear algebra calculus
and and series expansion and then also
there's the
appendices but these appendices are kind
of just like the the um the tip of the
iceberg uh a is kind of some of the code
but there could be a lot of work in
Matlab and other languages to make code
examples like that's what we're doing in
Julia right now appendix B goes through
the equations but still it's only
showing some and then here's oh c is the
code so a must be the
background yeah have you ever have you
ever watched oliv Oliver Nails uh
YouTube put a link I don't put a link I

don't recognize or no so here's the four
topics linear
algebra series
approximation like series expansions
function
approximation variational calculus and
stochastic Dynamic so systems that have
dynamical nature and probabilistic
nature but that's the
math that's what but this is also
foundational for going further into
basing mechanics because in many ways
it's just like more um remixes and and
kind of like concatenations
of these core mathematical
motifs there's been
[Music]
um there's been many people's
contributions to making the math more
um relatable it's just like it's a it's
a it's a great question to an area to
kind of run towards and contribute to
and explore like Jonathan shock is a
math
professor he wrote this gray text
um however it it may be like you know
more or less than someone is looking for
but it is really cool even like
like like even a P value or or um like a
bit there's like a lot of things to
return to so it's not like a check it
off your list because the basis of this
is just like surprise and
entropy so then that's like why it's
important to return to it many
times because it's not like you can just
cram it and finish the topic and then
just like never return to that

node here are 10
short
cool cool
yeah and then in the textbook group like
the work that uh octopus is doing right
now over the years people have brought
up many questions and thoughts about
learning math and about how Math
Connects and then octopus is now like
more um holding down these meetings and
just trying to coordinate more math
educational work but something that
that's always there always improvable if
somebody wants to start like just on any
part I think Axel made some models and
also mentioned that if people were
interested in that he'd be open to
collaboration
too octopus said that Axel Axel
constant um Axel Sorenson I think in the
chat oh Axel Sorenson okay cool yeah
cool yeah like for the math and stuff so
people could see the inter relationship
between the terms and things like
that yeah on
on on the
terms uh if people want to follow up
that like more formal but natural
language so not using equations per se
but
um going into like terms and in a
relationship that's like another
dimension with the active inference
ontology so that's the like that's the
generic system
description and then those are the terms
that actually apply to the generic
system so in certain ways it's simpler

than trying to give examples from
systems because it could be like oh the
action state of the robot well that's
like you know the the motor it's like
well but then the motor has a sense
element and then from the world's
perspective it's it is a you know so it
gets really specific when you are
dealing with specific systems but the
core
ontology is you don't even need to use
these 64 terms it's very small and the
kind of Kernel of it is the particular
partition and why and how there's a a
least action principle on a certain kind
of partitioned
graph with so all the equations just
setting up the path but then that's why
these kinds of um nodes and Edge figures
come up all the time because that is the
setting that's being
considered and that's coming more from
like the basic mechanics free energy
principle side whereas um in in the
textbook they actually set a target for
active
inference which is why in figure 1.2
active inference is is at the middle
between that free energy principle basy
mechanics and then the bottomup
construction of real math and
computational
models so this is more like applied math
and applied computer science because
it's like you're making matrices that
represent a certain system or certain
measurements and then up here is more
like the generic it's like like why does

gas burn why does wood do this and then
here's like this specific this
particular
engine so then that helps recognize like
somebody will make a claim analogously
to something like well because candles
burn my car will get 200 miles per
gallon it's like not
necessarily or because metal has some
structural Integrity this bridge design
is going to be fine
so the bottom and the low road is like a
very
empirical inquiry and
building and then here is just some of
the more generic elements that are
related to
physics so that's why this book brought
things together structured it in a
different way than like was
existing because it it um brings in that
low road High Road
approach which can be like dug into and
you can go deeper into low and high road
but just the layout of like kind of
setting context low in the high road
just one way to think about it but a
very useful one and then chapter four
when they come together and then chapter
five which is the most well-studied
domain at this time which is mammal
Neuroscience so that's why even though
there's sections like especially in four
that are just like whoa what is
happening um and and eventually we'll
have more of like generative dials like
turn up the math detail turn down the
math detail but um in a balanced form

going from here's the context then take
these two paths here's what it is here's
where it's being used it does have those
trails
um yeah the math the math learning
education Courtney thank you I'll copy
it in so like octopus can explore
it yeah PE people have made many
continua um what's kind of fun is
there's there's
hope for like doing something in a very
different way because if you think about
a technical concept Like A P value it's
hard enough to communicate technically
and basian um prior and posterior and
updating even before getting into it
just if they're learned one by one and
through the kind of Tech Tree Learning
trees that are how they're currently
taught it's going to a very long road
with a lot of attrition so I think
that's what motivates some of these
questions like about math learning that
people have um brought up over the
years that's a really interesting layer
to pull back
to because like computers are used
without math um needed by the user so as
that Continuum and that's doc starts to
materialize like what and how are people
learning what would be of epistemic and
pragmatic value for them all those kinds
of questions then San jev's textbook the
fundamentals uh which he has finished
the first volume of it's um two volumes
or this is just what I understand now
first volume is related to active
inference so that's kind of coming at it

from the low road largely second volume
is free energy principle phasing
mechanics however the the first volume
is extremely
comprehensive gives a
mega synthesis that's that will be um
comprehensive enough for like very
serious extended study and also use in
language models Etc um so that will be
very nice on the low road and then I
think that's why is because it gives a
more time for
codification and synthesis on the high
road
Elements which I mean even if there is
as uh minim I mean that's the whole
thing with Maxwell's equation and
Newton's equation is sometimes it IT
projects down to these Expressions that
are just so small
yeah I mean there's a lot of a lot of
there's a lot to go into
here I suppose I suppose that um
actually like typically in an act of
inference you're you're dealing with
equilibrium States but you're always
working it's always working to minimize
the free
energy so like it's it's non
equilibrium yes importantly it can
include that situation so like you could
have a um a cup of coffee in the room
and they're at thermal equilibrium but
then you have non-equilibrium
thermodynamics which is like Heating and
Cooling and more phenomena if it was
like a huge temperature difference or
something so similarly

it's good to check like yes I'm taking
in this one data point I'm updating the
prior here it is like and have a kind of
um like it's at equilibrium like the
temperature the thermometer setting 70
my estimate is 70 it should stay at 70
but then you can explore all of these
situations where that is
relaxed yes so you have like a Colman
alter which is like kind of a sliding
window memory then if you add it as just
one thermometer measurement it's going
to jerk around and just reflect the
measurements or if you had really like
too heavy it would just start taking
long run average so then that's like a
basian question like what should the
window and the forgetting be on the
common filter and then the whole thing
with the particular partition is like
now imagine if you were fitting that on
the flows of all four of
these so for each variable they're all
kind of undergoing that flow-based
unfolding which could be static or it
could be noisy or it could be um bodal
it's like trapped here and then it's
trapped here and then like what is the
structure of all those variables it's
not just for if you want to do something
more than a thermometer
abstraction but this is like the kernel
that describes the particular partition
and why and how
that's coming from um or why and how you
can have a path of least action on that
flow which is what is set up
here and and improved and clarified also

in but but this will always come up it's
like there's the four
states sense action internal and
external so fun and different letters
depending on different people's use and
so on um and then what their their time
development is plausibly a function
of and that's reflected by the
connections between the
nodes
um like that is what
this is showing which is that there
isn't a direct Edge between internal and
external that's tological
because it's what defines the interface
if there was a direct Edge you're just
talking about a different situation but
these don't necessarily map on so again
none of this is a claim about how things
are structured in the world it's like if
you are using a base graph anything
other than a fully connected graph so if
you have everything's connected to every
node then there's no partition it's just
like one connected Mass
um if there are pieces that aren't
connected to each
other then this situation comes into
play that that's interesting yeah
totally and it's like a it's like a
separate sort of um little
subgraph yeah exactly like the the the
um this is the niche and this is our
Roomba you know act the motors again
just simplifying the motors and the
internal States and the temperature
reading it's like we we have partitioned
the Roomba from the niche but if there

was like some sort of
wire between them inter is then it
wouldn't be internal States so it's like
that's the whole chapter six Plus+ plus
question how do you make these kinds of
maps for cognitive
systems and and that helps sometimes to
clarify like the the epistemic status of
these
Maps because like why is this reflected
that way it's like because it was
constructed that way as the
map but they they don't like just taking
alone support or reject some kind of way
that the world
is but making them within a certain um
type brings the generality of cognitive
modeling and then that's kind of just
like the
the how specific will you make the model
like is the engine just one
chamber and I just like I'm just looking
at heat across an engine overall or do
go and model Heat at like the components
or would do you model heat transfer with
this but those are all empirical and the
the person of the team building that
just addresses
those as they actually
happen how many chambers should we model
this engine is having it's like why are
we building the model who's it for what
questions are being asked that's what
chapter six goes into
um Jeff also this morning and just like
we're that was fun it was about the
inspiration in the
dream

but it was it it was fun and and uh but
the the pendulum is going to give more
that's going to come across more as like
a uh that's more like solving an
exercise in a physics class if it's like
we have an oscillator with two states
it's oscillating due to this write the
generative model that does this that's
kind of that I mean it has like a kind
of homework style feel not that there'd
be only one answer but there would be a
Continuum of problems from like there's
only one answer as stated to there's
multiple ways that you could have done
that on through I mean it was just
like where's time time perception
but then those those are those can you
can still play with it and and provide
and take different thoughts on it even
if it's not and then it just like oh
well it's not about generating one
specific model but if it was um if we're
a startup and it the the financial um
incentive was to make the thermometer
with this or that property then it' be
again just more of an engineering
question
also how it kind of
revisits like chapter 4 shows discreet
and
continuous in one
chapter like what's shared about them
and then seven and eight revisit
it that's like just showing even from
the beginning
like it's the situation choice to model
even how time
is okay okay

yeah I'll I'll put this in math learning
resources you you can add things um Adam
or
anyone like and if you're not sure yeah
that's collaborative
fun it's adding questions I mean the
hundreds of questions that have been
asked did not come from nowhere and
somebody with many questions could
contribute many questions
and someone who goes through many
resources could just say here's what
this one I just found it this way no
one's going to think it's like a total
final judgment just this was useful for
me learning this from
here at least it helps show what
different kinds of activity and
different like practices can
mean and it it just helps show examples
oh yeah these are ways to there's no um
death gap between like not knowing
linear algebra and slightly
understanding
it but then that that comes down to
everyone's specific
situation how much time and what kind of
thing and
like yeah graphs like the the learnable
loop that we this most recent blog
post where
kobus is able from RX and
fur for the a drone model for this to be
automatically
generated it's it's an incredible game
Cher it's it's amazing work it like Leap
Frogs what visualization capacity for
example exists in um python or um matlb

not not because it's like a new
visualization package it's just like you
would be kind of starting on Square zero
like okay I have a BAS graph represented
in mat labra and python but I'm I going
to just do like spread them all out or
how is this even going to work and for
it to Output like publication
ready and with a verified Integrity from
how it's actually specified not being
like this extensive PowerPoint um just
kind of offchain moves which is doing
doing one's best in the absence of this
but I mean this is like wow
so what
what what exactly is that that graph
figure there yeah great question so
there's there um
and like continue to dig and and explore
the papers in live stream where this is
explored more but basically briefly
there's there's two primary kinds of
graphs that come into play in um active
inference the first one is a basian
graph so in that situation the no are
random variables and the edges res um
resemble like causal influence so that's
like causal modeling um all those kinds
of like basian models from J Pearl and
others nodes are variables edges are
like functions that relate variables
their Maps it turns out over the last
like 10 years being applied more to to
like Bert devise and Carl friston but
the fundamentals kind of lying in the
pre previous decades every base graph
has also this dual representation as
what's called a forny

graph
yeah porny graphs basically the nodes
and the edges have the the opposite
character so it's like whereas in if in
a base graph you could have one node
with like 50 connections because it
could just be influenced by 50 variables
um whereas The Edge can only have two
nodes it's connected to the forny graph
kind of inverts that where the nodes are
like um edges in the base graph and vice
versa and it there's there's like a dual
relationship you can kind of represent
them both ways and and go between those
two but it turns out that in certain
forny graphs you can take the the basian
causal graph um you know probability of
rain probability of that here's the
observation so that's the kind of like
systems thinking how are things
connected to each other and then there's
this rendering that RX and fur does
under the hood to another structure
that's equivalent and does the same
calculation but it does it in this way
that can be like scheduled whereas on a
Bas graph there'd be a lot of Dees of
freedom in like how do you properly
update this and percolate messages and
the factor graph gives a much stronger
way to do that this is a technical
detail but it's like what is making the
RX infer method more oriented towards
the scaling and the engineering type
efficacy because it's it's a super
strong engine under the hood for running
these 40 Factor
graphs but also enables this more basy

and semantic layer possibly for people
to interact with it but to see it like
we've seen it
before um and flip it also into the
version that actually relates to how
like the different things
connect so just like new level of of um
ability to to look into the the kind of
base computational layers of these
models in in but not by just making like
custom
visualizations like here's one some
other large
one like for and
also time varying autoaggressive this
highlights which is so um Wise by kobis
like this isn't just about
cybernetics this is just doing basy and
statistics like on time series which
could be like temperature climate
markets Etc so it doesn't always need to
relate to more questions about like
embodiment or communication amongst
agents they can just be like this is the
graph and then that's just a
regression so it's not a theory about
agents anything specific or any specific
type of agent per se I
mean everything of course I'm just
saying saying informal takes but the
fact that Kus is showing no you can use
RX and fur to do these just linear
aggression Auto regression
no no perception action
here okay um well whatever uh I also I
updated the calendar event so that it
should say a little more clearly like
which piece is being discussed but um

yeah we'll just continue on

so any last comment or otherwise thanks

thanks lot

cool appreciate it farewell see you

later